Chapter 9

Non-aqueous media

Relative permittivity

Acid-base behavior in non-aqueous solvents

Liquid NH_3, HF,H_2SO_4, BrF_3, N_2O_4

Ionic liquids

Supercritical fluids

Categories of non-aqueous solvents

Protic Solvents

(HF, H_2SO_4, MeOH)

Aprotic Solvents

(N_2O_4, BrF_3)

Coordinating Solvents

(MeCN, Et_2O, Me_2CO)

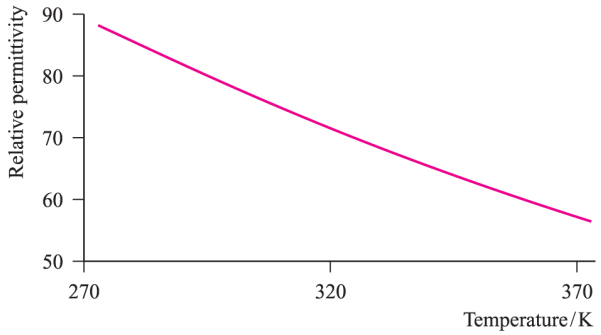
In Vacuum $$\text{Coulombic potential energy} = \frac{e^2}{4\pi\varepsilon_0 r} \quad (9.1)$$

In Media $$\text{Coulombic potential energy} = \frac{e^2}{4\pi\varepsilon_0\varepsilon_r r} \quad (9.2)$$

Solvent	Formula†	Relative permittivity, ε_r	Dipole moment, μ/debye
Formamide	$HC(O)NH_2$	109 (293 K)	3.73
Water	H_2O	78.7	1.85
Acetonitrile	MeCN	37.5 (293 K)	3.92
N,*N*-Dimethylformamide (DMF)	$HC(O)NMe_2$	36.7	3.86
Nitromethane	$MeNO_2$	35.9 (303 K)	3.46
Methanol	MeOH	32.7	1.70
Ethanol	EtOH	24.3	1.69
Dichloromethane	CH_2Cl_2	9.1 (293 K)	1.60
Tetrahydrofuran	C_4H_8O (structure **9.2**)	7.6	1.75
Diethyl ether	Et_2O	4.3 (293 K)	1.15
Benzene	C_6H_6	2.3	0

† Me = methyl; Et = ethyl.

Relative permittivity (dielectric constant) values at 298K (unless otherwise stated) for water and selected organic solvents.



Variation in the relative permittivity (dielectric constant) of water as a function of temperature.

Values of $\Delta_{transfer}H^o$ and $\Delta_{transfer}G^o$ for the transfer of ions from water to an organic solvent at 298 K

Ion	Methanol		Formamide		*N*,*N*-Dimethylformamide		Acetonitrile	
	$\Delta_{transfer}H^o$ / kJ mol^{-1}	$\Delta_{transfer}G^o$ / kJ mol^{-1}	$\Delta_{transfer}H^o$ / kJ mol^{-1}	$\Delta_{transfer}G^o$ / kJ mol^{-1}	$\Delta_{transfer}H^o$ / kJ mol^{-1}	$\Delta_{transfer}G^o$ / kJ mol^{-1}	$\Delta_{transfer}H^o$ / kJ mol^{-1}	$\Delta_{transfer}G^o$ / kJ mol^{-1}
F^-	12	20	20	25	–	≈60	–	71
Cl^-	8	13	4	14	18	48	19	42
Br^-	4	11	−1	11	1	36	8	31
I^-	−2	7	−7	7	−15	20	−8	17
Li^+	−22	4	−6	−10	−25	−10	–	25
Na^+	−20	8	−16	−8	−32	−10	−13	15
K^+	−19	10	−18	−4	−36	−10	−23	8
$[Ph_4As]^+$, $[BPh_4]^-$	−2	−23	−1	−24	−17	−38	−10	−33

Values of $\Delta_{transfer}H°$ and $\Delta_{transfer}G°$ for the transfer of ions from water to an organic solvent at 298 K.

Strengths of acids and bases

$$HX\ (aq) + H_2O\ (l) \rightleftharpoons H_3O^+\ (l) + X^-\ (aq)$$

Strength of acid HX depends on relative proton donor abilities of HX and H_3O^+

Tabulated values of K_a (or K_b) generally refer to the ionizations of acids in aqueous solution, thus when we consider HCl a strong acid in aqueous medium.

•Note, because of the **common ion effect**, if HCl is dissolved in acetic acid, the extent of ionization is less than in water and HCl acts as a weak acid.

•Non-aqueous solvents that are good proton acceptors (e.g. NH_3) enable acids to ionize.

•In a basic solvent, all acids are strong. The solvent is noted to exhibit a **levelling effect** on the acid, since the strength of the dissolved acid cannot exceed that of the protonated solvent.

•In aqueous solution, no acidic species can exist that is a stronger acid than H_3O^+.

‘Acids’ in acidic solvents

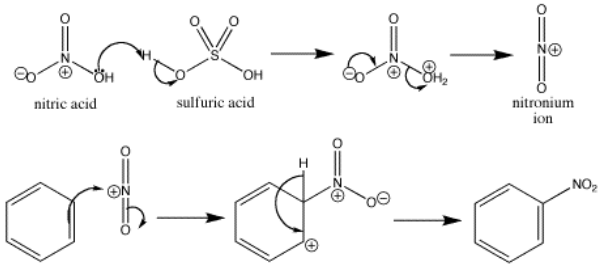
Effect of dissolving an ‘acid’ in acidic non-aqueous solvents enables us to realize that just because a compound is named an ‘acid’, it may not behave as one in non-aqueous media.

$$HNO_3 + 2H_2SO_4 \rightleftharpoons [NO_2]^+ + [H_3O]^+ + 2[HSO_4]^-$$

The above reaction may be described as the sum of:

$$HNO_3 + H_2SO_4 \rightleftharpoons [H_2NO_3]^+ + [HSO_4]^-$$
$$[H_2NO_3]^+ \rightleftharpoons [NO_2]^+ + H_2O$$
$$H_2O + H_2SO_4 \rightleftharpoons [H_3O]^+ + [HSO_4]^-$$

Formation of $[NO_2]^+$ is important in the nitration of aromatic compounds



Acids and bases: a solvent oriented definition

Self-ionizing solvent: an acid is a substance that produces the cation characteristic of the solvent, and a base is a substance that produces the anion characteristic of the solvent.

$2H_2O \rightleftharpoons [H_3O]^+ + [OH]^-$ (hydronium ion / hydroxide ion)

$2NH_3 \rightleftharpoons [NH_4]^+ + [NH_2]^-$ (ammonium ion / amide ion)

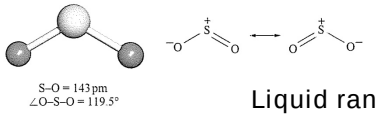
Non-aqueous solvents that are self-ionizing may be classified as:
1) proton containing (NH_3, HF, H_2SO_4, $HOSO_2F$)
2) aprotic (BrF_3, N_2O_4)

Dinitrogen tetroxide, N_2O_4, undergoes self-ionization.

$$N_2O_4 \rightleftharpoons [NO]^+ + [NO_3]^-$$

In this solvent medium, nitrosyl salts behave as acids and metal nitrates (MNO_3) act as bases.

Non-ionizing non-aqueous solvents

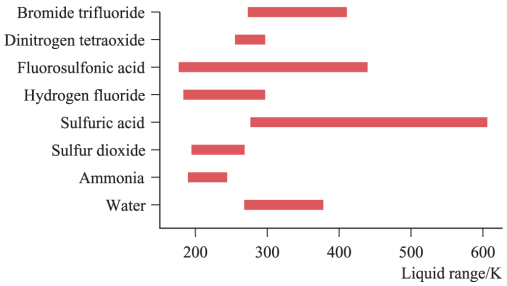


S–O = 143 pm
∠O–S–O = 119.5°

Physical properties of SO_2

Property / units	Value
Melting point / K	197.5
Boiling point / K	263.0
Density of liquid / g cm^{-3}	1.43
Dipole moment / D	1.63
Relative permittivity	17.6 (at boiling point)

Liquid ranges for water and selected non-aqueous solvents.



Liquid SO_2 is an inert solvent for both organic and inorganic compounds
- Good ionizing medium for compounds related to Ph_3CCl (giving $[Ph_3C]^+$)
- Useful in synthesis of group 16 and 17 cations species (e.g. I_3^+ and I_5^+)

$$3\ AsF_5 + 5I_2 \xrightarrow{\text{liquid } SO_2} 2[I_5][AsF_6] + AsF_3$$

Liquid ammonia

Property / units	NH_3	H_2O
Melting point / K	195.3	273.0
Boiling point / K	239.6	373.0
Density of liquid / g cm^{-3}	0.77	1.00
Dipole moment / D	1.47	1.85
Relative permittivity	25.0 (at melting point)	78.7 (at 298 K)
Self-ionization constant	5.1×10^{-27}	1.0×10^{-14}

- Relative permittivity of NH_3 (25) is lower than that of water (79), thus the ability of liquid ammonia to dissolve ionic compounds is less than that of H_2O.
  - Exceptions include $[NH_4]^+$ salts, iodides, and nitrates which are readily soluble

AgI, $K_{sp} = 8.3 \times 10^{-17}$, is sparingly soluble in water, but dissolves easily in liquid NH_3 (solubility = 206.8 g per 100 g of NH_3)

- The Ag^+ and I^- interact strongly with the solvent, and Ag^+ forms an ammine complex.

Precipitation reactions in liquid ammonia

In aqueous solution:

$BaCl_2\,(aq) + 2AgNO_3\,(aq) \rightarrow 2AgCl\,(s) + Ba(NO_3)_2\,(aq)$

In NH_3:

$Ba(NO_3)_2\,(solv) + 2AgCl\,(solv) \rightarrow BaCl_2\,(s) + 2AgNO_3\,(solv)$

Solubility of AgCl is 0.29 g/100 g liquid NH_3 compared with 1.91×10^{-4} g per 100 g H_2O

In NH_3:

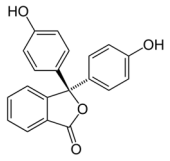
$KNO_3\,(solv) + AgCl\,(solv) \rightarrow KCl\,(s) + AgNO_3\,(solv)$

Neutralization reactions in liquid ammonia

Acid + Base → Salt + Ammonia

$NH_4Br + KNH_2 \rightarrow KBr + 2NH_3$

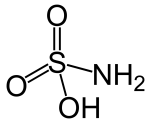
Phenolphthalein



Influence of solvent on acids

$H_2NSO_2OH\,(aq) + H_2O(l) \rightleftharpoons H_3O^+\,(aq) + [H_2NSO_2O]^-\,(aq)$

$K_a = 1.01 \times 10^{-1}$; a monoprotic acid in aqueous solution



sulfamic acid

$H_2NSO_2OH(aq) + 2KNH_2(l) \rightleftharpoons K_2[HNSO_2O] + 2NH_3\,(aq)$

a diprotic acid in liquid NH_3

The levelling effect of NH_3 means the strongest acid possible in this medium is NH_4^+, thus solutions of ammonium halides may be used as acids:

$Mg_2Si + 4NH_4Br \rightarrow SiH_4 + 2MgBr_2 + 4NH_3$

$Na_3As + 3NH_4Br \rightarrow AsH_3 + 3NaBr + 3NH_3$

Solutions of s-block metals in liquid NH_3

All Group 1 metals and group 2 metals Ca, Sr, and Ba dissolve in liquid NH_3.

Dilute solutions of the metals are bright blue, color arising from a broad and intense absorption in the IR region.

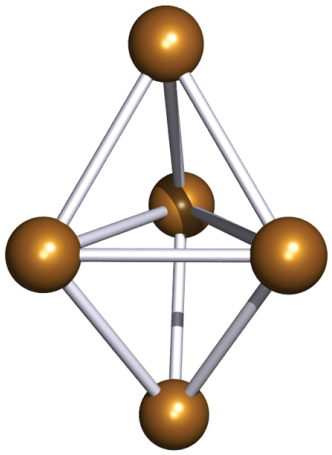
$$M \xrightarrow{\text{dissolve in liquid } NH_3} M^+(solv) + e^-(solv)$$

Dilute solutions occupy a volume greater than the sum of the metal plus solvent, with electrons occupying cavities of radius 300-400 pm.

- Dilute solutions are paramagnetic and the magnetic response corresponds to that of one free electron per metal atom.
- Molar conductivity initially decreases with increasing concentration, reaching a minimum near 0.05 M. Conductivity increases at higher concentrations and in saturated solutions the concentration is comparable with the solid metal.
- Saturated solutions are not blue and paramagnetic, but instead are bronze and diamagnetic.
  - Termed expanded metals and sometimes called an electride
- Decompose slowly $2NH_3 + 2e^- \rightarrow 2[NH_2]^- + H_2$
- Used as a *reducing agent* in a variety of reactions

Zintl ions

Early synthetic routes for Zintl ions involved reductions of Ge, Sn, or Pb in solutions of Na in liquid NH_3.



Zintl ion $[Sn_5]^{2-}$

$$Sn \xrightarrow{\text{Na in liquid } NH_3} NaSn_{1.0\text{-}1.7} \xrightarrow[\text{1,2-ethanediamine}]{\text{2,2,2-crypt in}} Na(crypt\text{-}222)_2[Sn_5]$$

Redox Reactions in liquid NH_3

Reduction half-equation	E^o / V in aqueous solution	E^o / V in liquid ammonia
$Li^+ + e^- \rightleftharpoons Li$	−3.04	−2.24
$K^+ + e^- \rightleftharpoons K$	−2.93	−1.98
$Na^+ + e^- \rightleftharpoons Na$	−2.71	−1.85
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76	−0.53
$2H^+ + 2e^- \rightleftharpoons H_2$ (g, 1 bar)	0.00	0.00
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.34	+0.43
$Ag^+ + e^- \rightleftharpoons Ag$	+0.80	+0.83

Selected standard reduction potentials (298 K) in aqueous and liquid ammonia media

- The oxidizing ability of each metal ion is solvent dependent.

Liquid HF

Corrodes glass containers, must be used in PTFE or dry Cu or Monel metal

$$4HF + SiO_2 \rightarrow SiF_4 + 2H_2O$$

Liquid range: 190 to 292.5 K
Relative permittivity 84 at 273 K; 175 at 200 K

Liquid HF undergoes self-ionization: $K_{self} = 2 \times 10^{-12}$ at 273 K

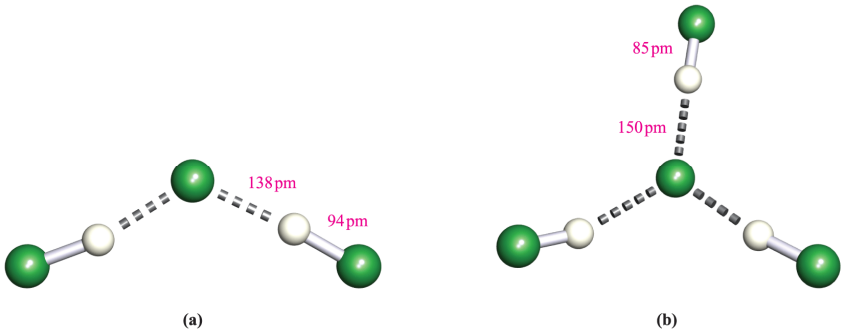
$$3HF \rightleftharpoons [H_2F]^+ + [HF_2]^-$$

dihydridofluorine(1+) ion — difluorohydrogenate(1-) ion

Large electronegativity difference between H (x^P = 2.2) and F (x^P = 4.0) results in the presence of extensive intermolecular hydrogen bonding in the liquid.

- Hydrogen bonded molecules (~7 molecules on average) in liquid phase
- Cyclic $(HF)_x$ species are present in the gas phase

Acid – Base behavior liquid HF



The structures of the anions (a) $[H_2F_3]^-$ and (b) $[H_3F_4]^-$, determined by low-temperature X-ray diffraction for the $[Me_4N]^+$ salts.

Sulfuric acid

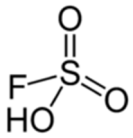
Property / units	Value
Melting point / K	283.4
Boiling point / K	≈603
Density of liquid / g cm^{-3}	1.84
Relative permittivity	110 (at 292 K)
Self-ionization constant	2.7×10^{-4} (at 298 K)

$2H_2SO_4 \rightleftharpoons [H_3SO_4]^+ + [HSO_4]^-$ $\quad K_{self} = 2.7x10^{-4}$

$2H_2SO_4 \rightleftharpoons [H_3O]^+ + [HS_2O_7]^-$ $\quad K_{self} = 5.1x10^{-5}$

Fluorosulfonic acid, HSO_3F

Property / units	Value
Melting point / K	185.7
Boiling point / K	438.5
Density of liquid / g cm^{-3}	1.74
Relative permittivity	120 (at 298 K)
Self-ionization constant	4.0×10^{-8} (at 298 K)



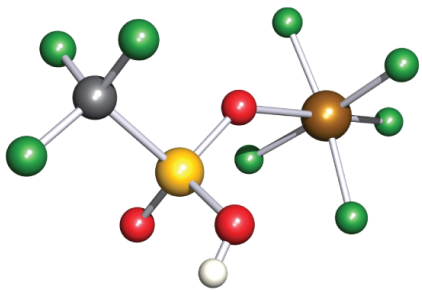
$$2HSO_3F \rightleftharpoons [H_2SO_3F]^+ + [SO_3F]^-$$

Superacids

Superacids, stronger acid than anhydrous H_2SO_4. Made by dissolving a strong Lewis acid (e.g. SbF_5) in either of the bronsted acids HF or HSO_3F capable of protonating even hydrocarbons, include mixtures of:

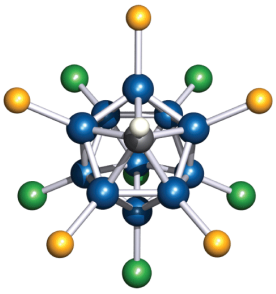
- HF and SbF_5
- HSO_3F and SbF_5 (called magic acid)

$$2HSO_3F + SbF_5 \rightleftharpoons [H_2SO_3F]^+ + [F_5SbOSO_2F]^-$$



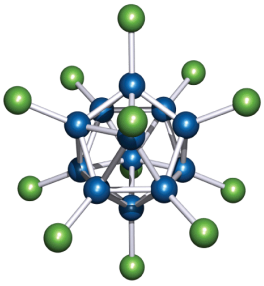
$SbF_5OSO(OH)CF_3$

Carborane Superacids



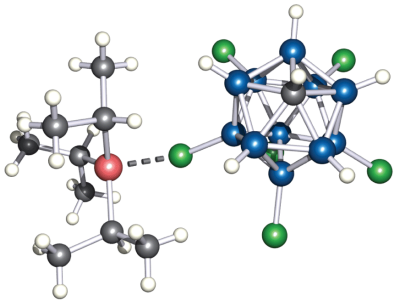
(a)

$[CHB_{11}R_5X_6]^-$(R = H, Me, Cl and X = Cl, Br, I)



(b)

$[B_{12}X_{12}]^{2-}$(X = Cl, Br)



(c)

$^{i}Pr_3Si(CHB_{11}H_5Cl_6)$

The long Si–Cl 'bond' which suggests that the structure approaches an ion-pair.

BrF_3

Good oxidizing and fluorinating agent

Property / units	Value
Melting point / K	281.8
Boiling point / K	408
Density of liquid / g cm^{-3}	2.49
Relative permittivity	107
Self-ionization constant	8.0×10^{-3} (at 281.8 K)

Ag + Sb –liquid BrF_3 → $Ag[SbF_6]$

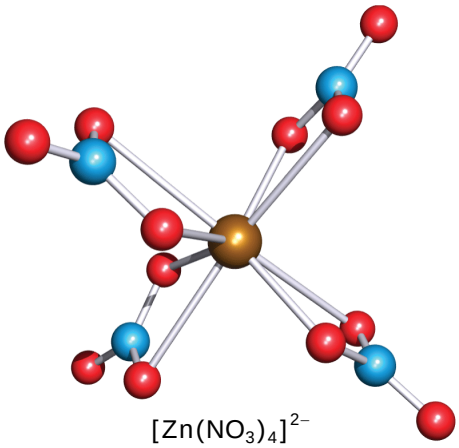
Ag + Au –liquid BrF_3 → $Ag[AuF_4]$

Dinitrogen tetroxide

Good oxidizing and nitrating agent

Property / units	Value
Melting point / K	261.8
Boiling point / K	294.2
Density of liquid / g cm^{-3}	1.49 (at 273 K)
Relative permittivity	2.42 (at 291 K)

$Li + N_2O_4 \rightarrow LiNO_3 + NO$



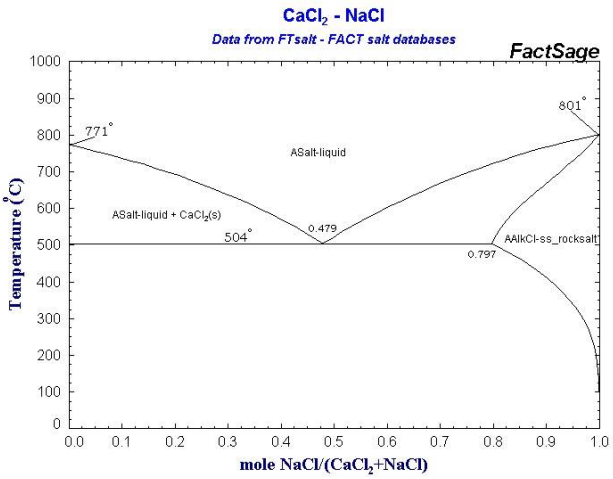
$[Zn(NO_3)_4]^{2-}$

Molten salt solvent systems

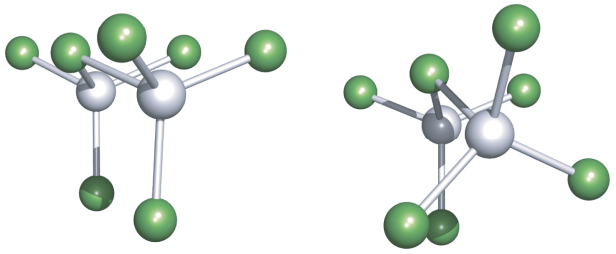
Molten salts: NaCl(*s*) –heat, T = 1073 K→ NaCl(*l*)

$2HgBr_2 \rightleftharpoons [HgBr]^+ + [HgBr_3]^-$

Consider the **eutectic**, a mixture of two substances that is characterized by a sharp melting point lower than that of either of the components. A eutectic acts as a single substance.



$NaCl\text{-}Al_2Cl_6$ mixture



(a) (b)

$Al_2Cl_6 + 2Cl^- \rightleftharpoons 2[AlCl_4]^-$

$2[AlCl_4]^- \rightleftharpoons [Al_2Cl_7]^- + Cl^-$

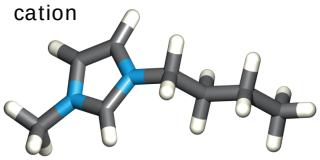
From $NaCl\text{-}Al_2Cl_6$ mixture (T_{melt} = 446 K)

Ionic Liquids

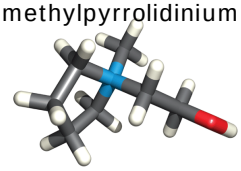
an ionic salt that is a liquid below 373 K

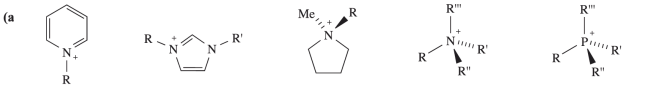
Commonly used ions in ionic liquids

1-butyl-3-methylimidazolium cation



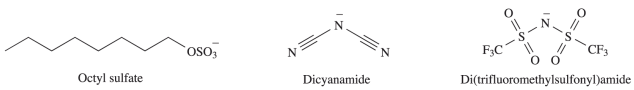
1-(2-hydroxyethyl)-1-methylpyrrolidinium cation



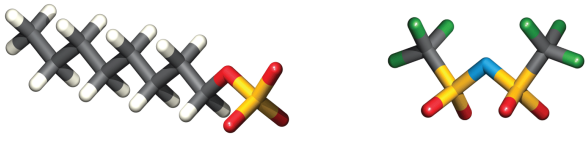


1-Alkylpyridinium ions; 1,3-Dialkylimidazolium ions; 1-Alkyl-1-methyl-pyrrolidinium ions; Tetraalkylammonium ions; Tetraalkylphosphonium ions

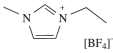
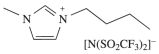
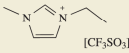
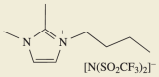
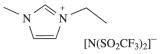
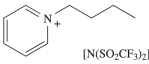
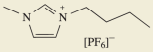
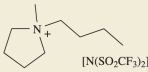
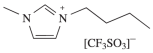
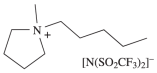
Acetate, $[AcO]^-$; Iodide, I^-; Hexafluoridophosphate, $[PF_6]^-$; Tetrafluoridoborate, $[BF_4]^-$; Sulfate, $[SO_4]^{2-}$; Trifluoromethanesulfonate (triflate), $[CF_3SO_3]^-$



Octyl sulfate; Dicyanamide; Di(trifluoromethylsulfonyl)amide

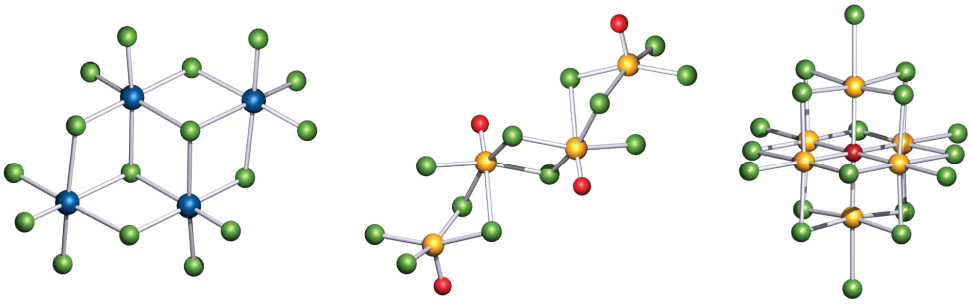


(c) (d)

Ionic liquid	$\Delta_{lattice}H^{\circ}$/kJmol^{-1}	Ionic liquid	$\Delta_{lattice}H^{\circ}$/kJmol^{-1}
	−492		−420
	−464		−412
	−425		−420
	−457		−414
	−449		−411

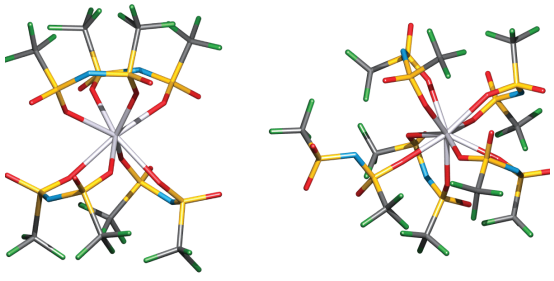
Data: I. Krossing *et al.* (2006) *J. Am. Chem. Soc.*, vol. 128, p. 13427; I. Krossing *et al.* (2007) *J. Am. Chem. Soc.*, vol. 129, p. 11296.

Calculated values of lattice enthalpies ($\Delta_{lattice}H^{\circ}$) for selected ionic liquids. Values refer to the process: gaseous ions → ionic solid



(a) (b) (c)

$[Bi_4Cl_{16}]^{4-}$ $[V_4O_4Cl_{12}]^{4-}$ $[Zr_6MnCl_{18}]^{5-}$

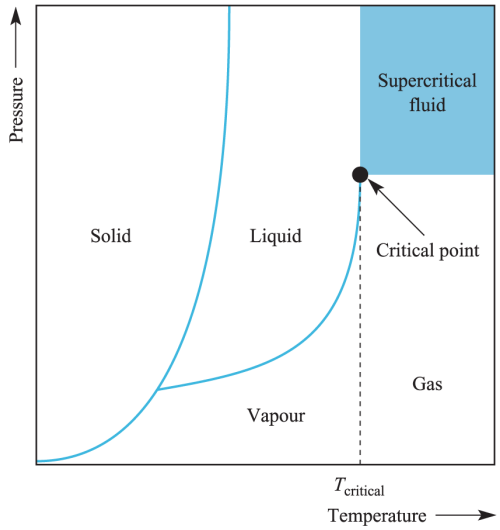


(a) (b)

$[Ca\{N(SO_2CF_3)_2\}_4]^{2-}$ $[Nd\{N(SO_2CF_3)_2\}_5]^{2-}$

Supercritical fluids

Pressure–temperature phase diagram for a one-component system.



Critical temperatures and pressures

Compound or element	Critical temperature / K	Critical pressure / MPa†
Xenon	289.8	5.12
Carbon dioxide	304.2	7.38
Ethane	305.4	4.88
Propane	369.8	4.25
Ammonia	405.6	11.28
Pentane	469.7	3.37
Ethanol	516.2	6.38
Toluene	591.8	4.11
1,2-Ethanediamine	593.0	6.27
Water	647.3	22.05

† To convert to bar, multiply by 10.

Primary commercial applications of supercritical fluids: coffee decaffeination, hops extraction

